

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – MOCK INTERVIEW (ANNEXURE A)

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Course Name: Computer Networks

Course Code: CSA07

Course Outcome: CO5 – Measure network performance using key metrics with simulation tools

Section A: Conceptual & Analytical Questions (Q1 – Q10)

Q1. Analyze how network congestion affects packet delay and packet loss.

A: As traffic approaches a link's capacity, router queues fill up, so packets wait longer before being forwarded – increasing delay. Once buffers overflow, incoming packets are dropped, causing packet loss. Both effects compound: TCP reacts to loss by retransmitting and shrinking its congestion window, which can further reduce throughput and, if congestion persists, worsen delay in a feedback loop.

Q2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

A: Star topology depends entirely on the central switch – if it fails, the whole network goes down, though individual host-link failures only affect that one host. Mesh topology provides multiple redundant paths between nodes, so it tolerates link or node failures far better, at the cost of much higher cabling and configuration complexity. Star is simpler and cheaper; mesh is more reliable but less scalable and more expensive.

Q3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically.

A: Each router adds processing delay (examining the header and making a forwarding decision), queuing delay (waiting behind other packets), and transmission delay (pushing the packet onto the next link) before the packet can even continue its journey. Since end-to-end delay is the sum of these delays at every hop plus propagation delay, adding more routers simply adds more of these per-hop delay components in series.

Q4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

A: QoS classifies and prioritizes real-time traffic (voice/video) over less time-sensitive data using mechanisms like traffic marking (DSCP), priority queuing, and bandwidth reservation. This ensures voice and video packets get forwarded with minimal queuing delay and jitter, and are less likely to be dropped during congestion, which is critical since these applications are highly sensitive to delay and jitter but can tolerate very little of either.

Q5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

A: Fiber optic cable offers much higher bandwidth, far greater maximum distances without a repeater, and immunity to electromagnetic interference compared to copper, making it ideal for backbone and high-speed links. The trade-offs are higher material and installation cost, the need for

specialized termination/splicing skills, and more fragile physical handling — so copper (e.g., Cat6) remains preferred for short, cost-sensitive desktop connections.

Q6. Analyze how network segmentation using VLANs improves performance and security.

A: VLANs logically divide a physical switch into separate broadcast domains, so broadcast traffic from one department or device group doesn't flood the entire network — reducing unnecessary traffic and improving performance. Security improves because devices on different VLANs cannot communicate directly without going through a router or firewall, allowing access control policies to be enforced between segments (e.g., isolating guest Wi-Fi from internal servers).

Q7. Compare the effects of TCP and UDP on network performance for different applications.

A: TCP provides reliable, ordered delivery with congestion control, which suits applications like file transfer, email, and web browsing where correctness matters more than speed — but this adds overhead and can increase latency under loss. UDP is connectionless and has no retransmission or ordering guarantees, giving lower latency and overhead, which suits real-time applications like VoIP, video streaming, and online gaming where a late packet is worse than a lost one.

Q8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability.

A: Load balancing distributes incoming requests or traffic across multiple servers or links instead of overloading a single resource, which improves response time and throughput and prevents any single point from becoming a bottleneck. It also improves availability, since if one server or path fails, traffic is automatically redirected to healthy resources, allowing the network/service to keep running with minimal disruption.

Q9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

A: Frequent routing updates help the network adapt quickly to topology changes, improving convergence time and avoiding stale/incorrect routes — but they also consume additional bandwidth and router CPU for processing, which can be significant in large networks. There's a trade-off: too-frequent updates waste resources and can even cause instability (route flapping), while too-infrequent updates slow convergence after a failure.

Q10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance.

A: Firewalls inspect and filter traffic, and encryption (e.g., TLS/IPsec) scrambles data for confidentiality — both add processing overhead and latency, since every packet must be examined or encrypted/decrypted. The trade-off is that stronger security generally means somewhat reduced throughput and higher latency, so organizations must balance acceptable risk against performance requirements, often using dedicated hardware (e.g., hardware-accelerated encryption) to minimize the impact.

Section B: Scenario-Based Problem Solving (Q11– Q20)

Q11. A network has high delay— how will you identify and fix the issue?

A: I would first measure where the delay occurs using tools like ping and traceroute to see which hop

introduces the largest jump in latency. Then I'd check for congestion (high link utilization), routing inefficiencies (too many hops), or a misconfigured/overloaded device at that hop. Fixes could include upgrading bandwidth, optimizing routing paths, applying QoS to prioritize critical traffic, or replacing an overloaded device.

Q12. If throughput is low despite high bandwidth, what could be the reasons?

A: Possible causes include a small TCP window size limiting how much data can be in flight, high packet loss causing frequent retransmissions, high latency (bandwidth-delay product not being fully utilized), duplex mismatches or faulty cabling causing errors, or a bottleneck at an intermediate device (e.g., an underpowered router or firewall) rather than the link itself.

Q13. How would you reduce packet loss in a congested network?

A: I would increase buffer/queue capacity or implement active queue management (e.g., RED) to avoid tail-drop bursts, apply QoS to prioritize critical traffic, add bandwidth or load-balance across multiple links, and identify and fix any faulty hardware or interfaces causing errors. Traffic shaping at the source to smooth out bursts can also help avoid overwhelming the network.

Q14. A VoIP call is breaking frequently— how would you troubleshoot?

A: I'd first check jitter, packet loss, and latency on the path using tools like ping and a jitter/VoIP-specific test, since VoIP is very sensitive to these. I'd verify QoS is properly configured to prioritize voice traffic (e.g., DSCP EF marking), check for bandwidth contention from other applications, and confirm the jitter buffer settings on the VoIP endpoints are adequate — adjusting them or the network configuration based on what I find.

Q15. Given a network with high jitter, suggest improvements.

A: I would implement QoS with a dedicated priority queue for real-time traffic, use traffic shaping to smooth out bursty flows, ensure sufficient bandwidth so queues don't build up unpredictably, and tune jitter buffers on receiving devices to absorb variable arrival times. Reducing the number of hops or using a more consistent path can also help minimize variability in delay.

Q16. How will you simulate a congested network in Packet Tracer?

A: I would build a topology with multiple end devices sending continuous or large traffic flows (e.g., using generate traffic or repeated pings/file transfers) toward a shared link or router with limited bandwidth, then reduce that link's bandwidth setting so it becomes the bottleneck. I could also add background traffic to saturate the link and observe increased delay, drops, and queue behavior using Packet Tracer's simulation mode.

Q17. If packets are delayed in a router, what steps will you take?

A: I'd check the router's CPU and interface utilization to see if it's overloaded, inspect queue depths and drop counters on the relevant interface, verify there isn't a routing loop or inefficient path causing extra processing, and check for misconfigured QoS policies that might be de-prioritizing the traffic. Depending on findings, I'd upgrade hardware, optimize the routing table, or adjust QoS settings.

Q18. How would you optimize a network for online gaming?

A: I would prioritize minimizing latency and jitter over raw bandwidth, since gaming needs fast, consistent round-trip times. This includes applying QoS to prioritize gaming traffic, using a wired

connection instead of Wi-Fi where possible, choosing game servers geographically close to reduce propagation delay, and avoiding unnecessary hops or heavy background traffic competing for bandwidth.

Q19. Design a small network and explain how you will measure its performance.

A: I'd design a simple star topology with a central switch connecting a router (for internet access), a server, and several client PCs, with the router providing DHCP and NAT. To measure performance, I'd use ping to measure latency and packet loss, a bandwidth/throughput test between hosts, and traceroute to check the path and hop count, comparing results against expected baselines to spot bottlenecks.

Q20. If RTT is very high, how will you diagnose the issue?

A: I'd run traceroute to see at which hop the round-trip time jumps significantly, which usually pinpoints where the delay is introduced — this could be a congested link, a distant/inefficient routing path, or a slow-responding device. I'd also check whether the destination itself is slow to respond versus the network path being the cause, and compare RTT at different times of day to check for congestion patterns.

Section C: Communication & Presentation (Q21– Q30)

Q21. Explain network performance to a non-technical person.

A: Think of a network like a highway system for data. Network performance is basically how smoothly and quickly information can travel from one place to another — how many lanes are available (bandwidth), how fast cars can go without traffic jams (delay), how consistent the travel time is (jitter), and whether any cars get lost along the way (packet loss). Good performance means fast, reliable, and consistent delivery.

Q22. Describe throughput using a real-life example.

A: Throughput is like the actual amount of water coming out of a tap versus the pipe's maximum rated capacity. Even if a pipe is rated for a high flow (bandwidth), the real flow you get (throughput) might be lower due to a partially closed valve, buildup inside the pipe, or low water pressure — similarly, actual network throughput is often less than the theoretical bandwidth due to congestion, overhead, or errors.

Q23. Present how packet loss affects video streaming.

A: When packets carrying video data are lost, the receiving device is missing pieces of the video stream, which can cause visible artifacts, frozen frames, or the video pausing to rebuffer while it waits for missing or corrected data. Streaming services often mitigate this using buffering, adaptive bitrate (lowering quality to reduce bandwidth needs), and error-concealment techniques, but higher loss generally means a visibly worse viewing experience.

Q24. Explain delay types clearly in under 1 minute.

A: There are four main types of delay: propagation delay, the time for a signal to physically travel across the medium; transmission delay, the time to push all the packet's bits onto the link; processing delay, the time a router takes to examine and decide how to forward a packet; and

queuing delay, the time a packet waits in a buffer before being sent, which grows with congestion. Total end-to-end delay is the sum of these at every hop along the path.

Q25. Describe Cisco Packet Tracer in a professional way.

A: Cisco Packet Tracer is a network simulation tool developed by Cisco that allows users to design, configure, and test network topologies in a virtual environment using realistic device models such as routers, switches, and end hosts. It's widely used for education and certification training because it lets learners practice configuration commands, observe packet flow in simulation mode, and troubleshoot network behavior without needing physical hardware.

Q26. Explain QoS in simple and technical terms.

A: Simply put, QoS is like giving priority lanes to important traffic — for example, letting a video call go ahead of a large file download so the call doesn't lag. Technically, QoS refers to a set of mechanisms (traffic classification, marking with DSCP/CoS values, queuing algorithms, and bandwidth reservation) that manage how network resources are allocated to different types of traffic based on their priority and requirements.

Q27. How would you present your simulation results to a client?

A: I would start with a brief summary of what was tested and why it matters to their business, then present key metrics (delay, throughput, jitter, packet loss) visually using clear charts rather than raw numbers or logs. I'd highlight any bottlenecks found, explain their real-world impact in plain language, and end with concrete, prioritized recommendations for improvement.

Q28. Explain jitter using a real-world analogy.

A: Jitter is like a bus that's supposed to arrive every 10 minutes, but sometimes comes in 5 minutes and sometimes in 15 — the average interval might still be 10 minutes, but the inconsistency makes it hard to plan around. In networking, jitter is this same variation in the arrival time of packets, which is especially disruptive for real-time applications like voice and video that expect a steady, predictable stream.

Q29. Describe how you tested network performance in a lab.

A: In a lab setting, I built a small topology in Packet Tracer/GNS3 with routers, switches, and end devices, then used ping to measure latency and packet loss between hosts, and simulated file transfers to observe throughput. I varied link bandwidth and introduced background traffic to simulate congestion, then compared performance metrics before and after applying changes like QoS configuration to evaluate their impact.

Q30. Explain the importance of performance metrics in one structured answer.

A: Performance metrics — such as delay, jitter, throughput, and packet loss — matter because they let us objectively measure how well a network is meeting user and application needs rather than relying on guesswork. They help identify bottlenecks before users complain, validate whether upgrades or configuration changes actually improved performance, and set clear, measurable service-level expectations (SLAs) for network reliability and quality.

Section D: HR & Behavioral Questions (Q31– Q40)

Q31. Introduce yourself and explain your knowledge of network performance.

A: I'm Vemula Jaswanth, an engineering student with a strong interest in computer networks. Through my coursework and lab work, I've studied core performance metrics like delay, jitter, throughput, and packet loss, and how protocols such as TCP, UDP, and DNS behave under varying network conditions. I've also worked hands-on with tools like Cisco Packet Tracer to design topologies and analyze how configuration choices affect real-world performance.

Q32. Why are you interested in networking and simulation tools?

A: I find networking fascinating because it's the invisible backbone behind almost every application we use, and small design decisions can have a huge impact on real-world performance and reliability. Simulation tools like Packet Tracer let me experiment safely with different topologies and configurations, see the immediate effects of my choices, and build a practical intuition for network behavior that complements the theory.

Q33. What makes you confident in analyzing network performance?

A: I'm confident because I've combined theoretical understanding of protocols and metrics with hands-on practice — designing topologies, simulating congestion, and interpreting metrics like RTT, jitter, and packet loss to diagnose real issues. This mix of concept and practice means I can approach a performance problem methodically: measure first, form a hypothesis, then test a fix.

Q34. Explain a concept you are very strong in (related to networking).

A: I'm particularly strong in understanding TCP versus UDP trade-offs and how they affect application performance. I can explain how TCP's reliability mechanisms — flow control, congestion control, and retransmission — provide guaranteed delivery at the cost of added latency, while UDP's lightweight, connectionless design suits real-time applications that prioritize speed over guaranteed delivery, and I can map this to real examples like video conferencing versus file downloads.

Q35. How would you handle a question you don't know?

A: I'd be honest that I'm not fully certain, rather than guessing confidently and risking a wrong answer. I would explain my reasoning process based on related concepts I do know, make a best-effort logical attempt at the answer, and note that I'd verify the specific detail with documentation or further research before relying on it in a real scenario.

Q36. Describe a situation where you solved a technical problem.

A: While working on a network simulation, I noticed that a supposedly high-bandwidth link was showing unexpectedly low throughput. I methodically checked for packet loss, verified the TCP window settings, and traced the path hop by hop, eventually identifying that a misconfigured duplex setting on one interface was causing errors and retransmissions. Correcting the duplex mismatch immediately restored expected throughput.

Q37. Why should we select you for a networking role?

A: I bring a solid theoretical foundation in networking combined with genuine hands-on experience simulating and troubleshooting real network scenarios, not just memorized definitions. I approach problems methodically — measuring before concluding — and I'm eager to keep learning, which

means I can grow quickly into more advanced networking responsibilities.

Q38. What challenges did you face while learning Packet Tracer?

A: Initially, understanding how simulation mode differs from realtime mode, and correctly interpreting how packets move hop-by-hop was challenging. I also found configuring more advanced features like VLANs and inter-VLAN routing tricky at first. I overcame these by building progressively more complex topologies, deliberately breaking configurations to see the effect, and cross-checking my understanding against documentation.

Q39. How do you improve your technical skills regularly?

A: I regularly practice by building and troubleshooting network topologies in simulators, reading up-to-date networking documentation and case studies, and working through practical assessments and coding challenges that connect theory to implementation. I also try to explain concepts to others, since teaching a topic quickly reveals any gaps in my own understanding.

Q40. Demonstrate confidence while explaining a complex concept simply.

A: Take TCP congestion control: essentially, TCP starts sending data cautiously, then gradually speeds up as long as packets arrive successfully — much like a driver slowly accelerating on an open road. The moment it detects a problem, such as a lost packet signaling congestion ahead, it slows down sharply and then carefully speeds back up again. This constant balancing act is what allows many devices to fairly share the same network without overwhelming it.